

Software Test Documentation

Skin Cancer Detection App Using Machine Learning

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1.0 Introduction

1.1 Purpose

This test plan's objectives are to specify the testing methodology for the Skin Cancer Detection App Using Machine Learning and confirm that the system meets the functional and non-functional requirements listed in the Requirements Document. To assess the system's anticipated behavior under both typical and atypical scenarios, the Test Plan defines test cases. The feature or behavior being tested, the necessary inputs, the anticipated outcomes, and the steps required to confirm proper system functionality will all be specified in each test case. Testing will be performed to identify potential errors and verify that the implemented system meets the project's specifications.

1.2 Scope

The functional and non-functional requirements of the Skin Cancer Detection App Using Machine Learning are tested in this test plan. The main features of the system, such as user authentication and guest access, image capture and upload, image preprocessing, communication with the machine learning service, benign or malignant prediction with a confidence score, storage and retrieval of analysis history for authenticated users, and deletion of stored user data, will all be tested. Relevant non-functional requirements, such as security, performance, usability, reliability, and appropriate handling of invalid or unexpected inputs, will also be assessed by the plan. To ensure that the implemented system operates as intended under both expected and unexpected conditions, each requirement will be linked to one or more test cases.

2.0 Test Items and Features

2.1 Features To Be Tested

- **Image Capture and Upload:** Testing the application's ability to capture a new skin lesion image using the device's camera or select an existing image from the device.
- **Image Preprocessing:** Testing the ability of the image preprocessing to load, resize, normalize, and process the skin lesion images before feeding them into the CNN.
- **CNN Model Accuracy:** Testing the ability of the model to classify the skin lesion images accurately.
- **Machine Learning Model Training:** Testing whether the prepared skin lesion dataset can be used to effectively train the chosen machine learning models and whether the training procedure results in a usable trained model.
- **Skin Lesion Prediction:** Evaluating the trained model's ability to classify a skin lesion image as benign or malignant.
- **Prediction Confidence Visualization:** Verifying if the program shows the confidence score related to the benign or malignant prediction.
- **Comparison with Training Data:** As stated in the requirements document, testing the system's ability to compare analytical results with training data.
- **Model Evaluation:** Testing the performance of the model in terms of accuracy and precision.
- **Application Interface:** Testing the application interface, buttons, and overall functionality.
- **Safety Information and Disclaimers:** Verifying whether the program shows the necessary medical disclaimers and safety information to let users know that the outcome is not a diagnosis.

- **User Authentication and Guest Access:** Testing guest access, login, logout, and account registration, including whether or not guests can utilize the image analysis features without registering.
- **History and Lesion Tracking:** Verifying the accessibility of previously saved skin lesion images and analysis results, including classification, confidence score, and date, for authenticated users.
- **Data Management:** Verifying the storage, retrieval, and deletion of user data, analysis records, and skin lesion photos in accordance with system requirements.
- **Client-Server Communication:** To submit images for analysis and retrieve the prediction and confidence score, the mobile application must be able to interface with the server/API.
- **Performance:** Evaluating whether key application functions, such as analysis and interface response times, meet the requirements.
- **Reliability:** Evaluating how well the application handles errors and unforeseen problems, such as when the server is unreachable.
- **Security:** Testing secure communication, access controls, authentication, and the preservation of user data and images of skin lesions.
- **Usability:** Testing whether the user can access and utilize the main functionalities of the application through the Android user interface.
- **Scalability and Maintainability:** Testing system features to support the growth and maintainability of the application.

3.0 Test Cases

3.1 Functional Requirement Test Cases

- Test Cases for FR-1: Input of an Image
 - Objective: Verify that the application allows the user to provide a skin lesion image through capturing a new image through the device's camera or selecting an existing image from the device.

Test Case ID	Test Case	Input/Action	Expected Result
FR1 - TC1	Capture a skin lesion image using the camera	The user chooses to capture an image through the camera	The application opens the camera and allows the user to capture an image.
FR1 - TC2	Select an existing image	User chooses to select an image	The application lets users select and

		from the device's photo library	upload an image.
FR1 - TC3	Camera permission granted	User chooses the capture option and grants camera permission	The camera opens, and the user can capture an image.
FR1 - TC4	Camera permission denied	User chooses the capture option and denies camera permission	The camera does not open and handles the denied permission without an error.
FR1 - TC5	Cancel image selection	User opens the photo library but cancels without selecting an image	The application returns to the previous page without submitting an image.

- Test Cases for FR-2: Image Preprocessing
 - Objective: Verifying that the system can preprocess the submitted skin lesion images through resizing, normalization, and preparing them for the analysis of the machine learning model.

Test Case ID	Test Case	Input/Action	Expected Result
FR2 - TC1	Preprocess a valid lesion image	The user submits a clear skin lesion image	The image is successfully preprocessed and prepared for the ML model.
FR2 - TC2	Resizing an image	User submits an image with different dimensions from the expected dimensions for the model	The image is resized for analysis.
FR2 - TC3	Normalize an image	User submits a valid image	The image pixel values are normalized for analysis.

FR2 - TC4	Process a blurry image	User submits an image that is too blurry for a reliable analysis	The image does not get processed, and the user is asked to submit another image.
FR2 - TC5	Process an unsuitable image	User submits an image that does not satisfy the application's image requirements	The image does not satisfy the requirements and was not sent for analysis.

- Test Cases for FR-3: Machine Learning Model Training
 - Objective: Verifying that the machine learning model is trained successfully with the provided skin lesion dataset and can be used for evaluation and inference.

Test Case ID	Test Case	Input/Action	Expected Result
FR3 - TC1	Train the selected model	Train the selected model using the skin lesion dataset	Model completes training and produces a trained model.
FR3 - TC2	Verifying training	Training the model with the specified number of epochs	Training and validation performance metrics are saved.
FR3 - TC3	Save the trained model	Complete training and save the model	The trained model is saved and can be used for evaluation or inference.
FR3 - TC4	Train with augmented data	Apply the selected augmentation techniques to the dataset and train the model	Training is done using the augmented dataset without failure.
FR3 - TC5	Handling of invalid training data	Start training when the data cannot be loaded	Training stops, returns an error.

- Test Cases for FR-4: Machine Learning Model Evaluation

- Objective: Verify that the trained model can be evaluated using the specified metrics to assess its image classification performance.

Test Case ID	Test Case	Input/Action	Expected Result
FR4 - TC1	Evaluating the model on test data	Run the trained model on the test dataset	The model successfully processes the test dataset and produces predictions without an error.
FR4 - TC2	Calculate accuracy	Comparing the produced predictions with the true labels	The system successfully calculates the model's classification accuracy.
FR4 - TC3	Calculate precision and recall/sensitivity	Evaluate predictions using true class labels	The system successfully calculates precision, recall/sensitivity.
FR4 - TC4	Calculate F1 score	Evaluate predictions using true class labels	The system successfully calculates the model's F1 score.
FR4 - TC5	Generate a confusion matrix	Evaluate predictions using true class labels	The confusion matrix is generated successfully, showing correct and incorrect classifications.
FR4 - TC6	Examine false negatives	Evaluate the cases where malignant lesions are classified as benign	False-negative predictions are identified from previously calculated evaluation metrics.
FR5 - TC7	Evaluate the model with unseen data	Provide test images that were not	The trained model produces

		provided during training	predictions for unseen images, showing how well it generalizes.
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- Test Cases for FR-5: Skin Lesion Prediction
 - Objective: Verify that the system analyzes processed images and the trained model returns a prediction.

Test Case ID	Test Case	Input/Action	Expected Result
FR5 - TC1	Predict a benign lesion	Submit a valid image whose prediction should be benign	The system returns a prediction.
FR5 - TC2	Predict a malignant lesion	Submit a valid image whose prediction should be malignant	The system returns a prediction.
FR5 - TC3	Predict an unseen image	Submit a valid image that was not used during training	The system returns a prediction.
FR5 - TC4	Reject an image before prediction	Submit an image that fails to satisfy the preprocessing requirements	The user is asked to provide another image.
FR5 - TC5	Handle unavailable ML service	Submit an image when the remote ML server cannot be reached	The application returns an appropriate server error message.

- Test Cases for FR-6: Prediction Confidence Visualization
 - Objective: Verify that the application displays the confidence score associated with the produced prediction in an understandable manner.

Test Case ID	Test Case	Input/Action	Expected Result
FR6 - TC1	Display confidence for a benign lesion	Submit a valid image whose	The system returns the benign

		prediction is benign	prediction with its corresponding confidence score.
FR6 - TC2	Display confidence for a malignant lesion	Submit a valid image whose prediction is malignant	The system returns the malignant prediction with its corresponding confidence score.
FR6 - TC3	Display confidence as a percentage.	Complete a successful analysis	The system's confidence score is presented to the user as a percentage.
FR6 - TC4	Display low-confidence prediction.	Submit an image that produces a low confidence score	The application shows the confidence score and the predictions, providing an appropriate safety guidance.
FR6 - TC5	Don't provide a confidence score when analysis fails.	Analysis fails (e.g., server cannot be reached)	The application returns an appropriate error message without displaying a prediction result or a confidence score.

- Test Cases for FR-7: Comparison with Training Data
 - Objective: Verify that the model can be tested using the skin lesion dataset and its predictions can be compared with the known labels of the images to evaluate classification performance.

Test Case ID	Test Case	Input/Action	Expected Result
FR7 - TC1	Compare the prediction with the known image label	Provide a labeled image to the model.	The model produces a prediction that can be compared with the class label.

FR7 - TC2	Identify a correct classification.	Compare a model that produces the same classification as the class label.	The prediction is identified as correct during evaluation.
FR7 - TC3	Identify an incorrect classification.	Compare a model's incorrect classification with the class label.	The prediction is identified as incorrect during evaluation.
FR7 - TC4	Compare the candidate model's performance.	Evaluate the candidate models using the evaluation metrics.	The models' performances are compared using the evaluation metrics.
FR7 - TC5	Evaluate previously unseen images.	Run the trained model with testing (unseen) data.	Understand how well the model can generalize.

- Test Cases for FR-8: Safety Information and Disclaimers
 - Objective: Verify that the application displays the necessary medical disclaimers stating that the provided prediction is not a medical diagnosis and is only intended for informational and awareness purposes.

Test Case ID	Test Case	Input/Action	Expected Result
FR8 - TC1	Display a medical disclaimer.	Open the application screen that displays safety information.	The application displays a medical disclaimer stating that the prediction is not a medical diagnosis.
FR8 - TC2	Display safety information with the analysis result.	Complete analysis of a skin lesion image.	The application displays safety information along with the prediction result.
FR8 - TC3	Safety guidance for high-confidence malignant results.	Produce a high-confidence malignant result.	The application displays safety information encouraging the user to seek a professional

			opinion, along with the prediction result.
FR8 - TC4	Safety guidance for low-confidence benign results.	Produce a low-confidence benign result.	The application displays safety information encouraging the user to seek a professional opinion, along with the prediction result.
FR8 - TC5	Access safety information	Open the application's information screen.	The information screen provides information regarding safety, limitations, and medical disclaimers.

- Test Cases for FR-9: History and Lesion Tracking
 - Objective: Verify that authenticated users can access their analysis history, which contains their skin lesion images and prediction results, in chronological order, while guest users don't have access to their history.

Test Case ID	Test Case	Input/Action	Expected Result
FR9 - TC1	Save analysis to history.	An authenticated user completes an analysis.	The completed analysis is saved to the authenticated user's analysis history.
FR9 - TC2	View analysis history.	An authenticated user selects to view their history.	The application displays an authenticated user's analysis history.
FR9 - TC3	Display analysis history.	An authenticated user views an entry from the history.	An entry in the history is displayed showing the analyzed image, classification result,

			confidence score, and timestamp.
FR9 - TC4	Display the history in chronological order.	An authenticated user opens the history containing one or more entries.	The application displays past entries in chronological order.
FR9 - TC5	Prevent guest users from having a history.	A guest user completes an analysis and exits the guest session.	A guest user's analysis results are discarded after the session ends.
FR9 - TC6	Prevent access to another user's history.	An authenticated user attempts to access another user's history.	The application permits a user to access another user's analysis history.

- Test Cases for FR-10: Data Management
 - Objective: Verify that the system correctly stores, retrieves, and deletes authenticated users' account information, analysis results, and corresponding images, while guest users don't have access to this feature.

Test Case ID	Test Case	Input/Action	Expected Result
FR10 - TC1	Store an authenticated user's analysis	An authenticated user successfully completes an analysis	The analysis result and its corresponding image are linked to the user's account.
FR10 - TC2	Retrieve an authenticated user's analysis result	An authenticated user requests to access a previous analysis	The analysis result and its corresponding image are successfully retrieved.
FR10 - TC3	Delete a stored analysis	An authenticated user requests to select and delete a history entry	The selected analysis result and its corresponding image have been successfully deleted.

FR10 - TC4	Delete a user account	An authenticated user requests to delete their account	The user's account information, all analysis results, and all submitted images are permanently deleted.
FR10 - TC5	Prevent saving a guest user's data	A guest user completes an analysis	The guest user's analysis result and submitted image are deleted after the session ends.
FR10 - TC6	Verify that the analysis is linked to the correct image	Retrieve a stored analysis result with its corresponding image	The correct image, along with the correct analysis record, is retrieved via an image reference.
FR10 - TC7	Prevent unauthorized data access	A user attempts to access data related to another user account	The system denies access to another account's data.

- Test Cases for FR-11: Application Interface
 - Objective: Verify that the application correctly transitions between screens, selects buttons, provides correct navigation, and allows users to access the application's features through the Android user interface.

Test Case ID	Test Case	Input/Action	Expected Result
FR11 - TC1	The application opens a login screen when opened successfully	Launch the application on an Android device.	The application opens successfully and displays the login screen.
FR11 - TC2	The application will request username and password information.	Enter the username and the password in the correct fields.	The application accepts the correct username and password in the correct fields.
FR11 - TC3	The application	Enter the username	The user is

	logs in if the username exists and the password is correct.	and its correct password, then click the Login button.	authenticated, and the home screen opens after successful login.
FR11 - TC4	The home screen displays the correct buttons.	Successfully log in or choose to stay as a guest.	The home screen is shown with the correct buttons and available navigation.
FR11 - TC5	The capture/upload image button will open a page that asks for permission to access the photo library or the camera.	Select the capture/upload image option.	The capture/upload image screen opens, asking permission to access the camera or the photo library.
FR11 - TC6	Access image history.	An authenticated user selects the history option.	The application opens the history page, displaying all previous analysis entries.
FR11 - TC7	Access the information screen.	Select the information option.	The application opens the information screen containing the application information, safety information, and medical disclaimers.
FR11 - TC8	Display analysis results screen.	Successfully submit an image and analyze the skin lesion.	The application displays the analysis result containing the classification prediction and the confidence score.
FR11 - TC9	Navigate back from a screen.	Select the back option from the application screen.	The application returns to the previous page without crashing.

FR11 - TC10	Access the application as a guest.	Choose the continue as a guest option on the login screen.	The application allows a guest user to use a one-time analysis feature without saving their data or requiring a login.
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3.2 Non-Functional Requirement Test Cases

- Test Cases for NFR-1: Performance

Objective: The program will execute its operations effectively while being able to answer user commands within specified time frames.

Test Case ID	Test Case	Input/Action	Expected Result
NFR1	Test application performance	Perform app navigation and image analysis	App responds within specified time frame limits

- Test Cases for NFR-2: Scalability

Objective: The application must remain efficient as more users and images are added to the system.

Test Case ID	Test Case	Input/Action	Expected Result
NFR2	Test application scalability	Increase number of created accounts and stored images	App maintains good performance without exceptional slowdowns

- Test Cases for NFR-3: Accuracy

Objective: The CNN shall make accurate predictions about the skin lesion type

Test Case ID	Test Case	Input/Action	Expected Result
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NFR3	Test prediction accuracy	Analyze test images with known classifications	The CNN produces consistent predictions with the expected classifications
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- Test Cases for NFR-4: Reliability

Objectives; The application consistently performs all its tasks without crashing

Test Case ID	Test Case	Input/Action	Expected Result
NFR4	Application reliability test	Perform multiple image analysis tasks through the app	App successfully completes operations without unexpected crashes

- Test Cases for NFR-5: Usability

Objectives: The application shall provide a user-friendly interface that makes it easy for non-technical users to navigate through

Test Case ID	Test Case	Input/Action	Expected Result
NFR5	App usability test	Analyze multiple scenarios in which the user needs to navigate to a destination	All scenarios are achieved with great ease

- Test Cases for NFR-6: Security

Objective: Application successfully protects user accounts and personal information

Test Case ID	Test Case	Input/Action	Expected Result
NFR6	App security test	Attempt to access	Unauthorized

		other users data without authorization	access is denied
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4.0 Conclusion

This document establishes the key features and the functional and non-functional requirements necessary for the app to operate as intended and efficiently. These requirements provide a foundation for developing test cases, which serve as a step-by-step process to test if each objective is being accomplished.